Kaidi **Zha**

UNDERGRADUATE RESEARCHER · AI & MACHINE LEARNING

Department of Physics, Tsinghua University, Beijing, China

(+86) 14790518086 | ckd23@mails.tsinghua.edu.cn | illusionsin30 | kaidi-zha

"Building intelligent systems that see, learn, and act."

Research Interests

Generative Models, Inference Acceleration

Dedicated to making large generative models faster and cheaper to deploy through **quantization and mixed-precision inference**, **efficient architecture design**, and **data-centric strategies**, with a focus on **inference throughput**, **memory efficiency**, and **training efficiency**.

Education

Tsinghua University — *Beijing, China*

B.S. IN MATHEMATICS AND PHYSICS — *Sept. 2023 - Present (Expected 2027)*

- Overall GPA: 3.6/4.0, with a steady upward trend — 3.9/4.0 in the most recent semester (Spring 2026).
- Earned A (4.0) in Pattern Recognition and Machine Learning, Reinforcement Learning and Control, and Fundamentals of Computer Graphics.

RELEVANT COURSEWORK

AI & ML	Pattern Recognition and Machine Learning (A), Reinforcement Learning and Control (A), Fundamentals of Computer Graphics (A), Statistical Machine Learning (B+)
Mathematics	Probability and Statistics (A), Experiments in Mathematics (A), Applied Stochastic Processes (A-), Complex Analysis (B+)
Computer Science	Data Structure and Algorithm (A-), Programming Fundamentals (A-), Modern Operating Systems (P), C++ Programming for Linux (P)

Research Experience

Q-DiT: Accurate Post-Training Quantization for Diffusion Transformers — *Supervised by Prof. Yuan Meng, Dept. of CS & Tech, Tsinghua University*

RESEARCH ASSISTANT — *Jun. 2025 - Present*

- **Method**: Co-designed a timestep-aware, layer-wise mixed-precision PTQ framework that allocates bit-widths (3/4/5-bit) per denoising timestep and per layer, driven by a three-way sensitivity metric (Hessian trace, GPTQ reconstruction error, Fisher information) and solved by dynamic programming under a global W4A4 budget.
- **System**: Refactored the codebase into *qdiffusers*, a modular YAML-driven quantization–evaluation pipeline; adapted it to **7 DiT-family models** (DiT-XL/2, FLUX.2 4B/9B/12B, PixArt-Σ, OpenSora, Qwen-Image, HunyuanVideo, Wan2.2) with multi-GPU parallel GPTQ support.
- **Results**: Led all ImageNet-1K experiments — W4A4 achieves near-lossless FID 7.76 on DiT-XL/2 256×256; on FLUX.2-klein-4B, W4A4 FID 25.22 even **surpasses the FP16 baseline** (26.08); OpenSora W4A8 drops VBench score by only 0.16%.

ArchEGraph: A Large-Scale Graph Dataset for Geometry–Topology–Physics Aligned Building Energy Modeling — *Supervised by Prof. Borong Lin, School of Architecture, Tsinghua Univ.; with UC Berkeley & MIT*

RESEARCH ASSISTANT — *Apr. 2026 - Present*

- **Role**: Second author of the manuscript and main contributor to the benchmark experiments of a dataset spanning **5,481 buildings and 49,326 validated simulation cases** ($>1.33\times10^5$ space nodes, 1.44×10^6 face nodes).
- **Contributions**: Designed and ran cross-building and cross-climate generalization benchmarks over 7 model families (XGBoost, DeepSets, Set Transformer, Perceiver, GATv2, TransformerConv, GraphGPS) to validate dataset quality on graph-reconstruction and load-prediction tasks.

Selected Projects

TrainYourOwnGPT: Building & Analyzing GPT-Style Language Models from Scratch
Pattern Recognition & Machine Learning, Course Project

TEAM OF 2; LED SCALING-LAW AND FINE-TUNING STUDIES *Spring 2026*

- Implemented a GPT-style decoder (RoPE, SwiGLU, pre-norm blocks) from scratch in PyTorch, reaching **105.15 validation PPL** on Penn Treebank; ran a 5-configuration hyperparameter sweep over 3 seeds.
- Conducted a scaling-law study across 0.77M–5.46M parameters and controlled ablations of QK normalization, attention gating, and value embeddings.
- Fine-tuned Qwen2.5-0.5B with LoRA (0.89% trainable parameters) on arXiv abstracts with a pre-registered quantitative evaluation of format shift; compared autoregressive vs. masked-diffusion LMs (MDLM, LLaDA), including A100 throughput/latency benchmarks.

fair-SPLIT: Near-Optimal Decision Trees for Accuracy, Sparsity, and Fairness
Statistical Machine Learning, Course Project

TEAM OF 2; DEVELOPED THE SPLIT-FAMILY METHODS AND EXPERIMENTAL DESIGN *Spring 2026*

- Implemented the SPLIT family of near-optimal sparse decision trees (dynamic-programming lookahead + greedy/optimal leaf completion) and benchmarked against CART on **12 datasets** × **5 seeds**, winning on COMPAS, HELOC, Heart, and Law School.
- Proposed Leaf-Pareto Fair Recalibration (LPFR), a deployable leaf-level post-processing method; label-aware calibration cuts the demographic-parity gap from 0.17 to **0.003** on Adult.
- Open-sourced at github.com/illusionsin30/fair-SPLIT.

AIDER: Taming Aleatoric Impulse in Off-Policy Reinforcement Learning
Reinforcement Learning & Control, Course Project

TEAM OF 8; EXPLORED DIFFUSION/FLOW-BASED MULTIMODAL POLICY EXTENSION *Spring 2026*

- Team built DEAR, an EMA-anchored evidential regression operator for end-to-end aleatoric uncertainty estimation, and AIDER, which damps value-overestimation impulses without heuristic annealing — best average performance on the DeepMind Control Suite among 7 model-free baselines.
- Deployed on a 4WD vehicle longitudinal speed-tracking task with **RMSE 0.349 m/s** (∼1.5% tracking error); personally explored integrating the framework with diffusion / normalizing-flow policies for multimodal action distributions.

CPU Path Tracer: A Physically-Based Renderer in C++
Fundamentals of Computer Graphics, Course Project

INDIVIDUAL PROJECT *Spring 2026*

- Built a renderer from scratch: Whitted ray tracing and Monte Carlo path tracing with next-event estimation, Cook–Torrance glossy BRDF, Schlick Fresnel, depth of field, and BVH acceleration.
- Engineered deterministic lock-free multithreading (`std::thread`) with a **49× speedup on 64 threads** and byte-identical output to the serial renderer.

Skills

Programming	Python, C/C++, CUDA, LaTeX
Deep Learning	PyTorch, JAX, HuggingFace Transformers & Diffusers, PEFT/LoRA, quantization toolchains
Tools	Git, Linux, Docker, multi-GPU training, OpenMP / `std::thread`
Languages	Chinese (Native), English (CET-4: 596, CET-6: 534)